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1 Step 08 — Safe Exploitation in Imperfect-Information Games

Games: Kuhn Poker (3-card, 2-player) and Leduc Hold'em (6-card, 2-round) — the smallest exact-solvable imperfect-information testbeds, reusing the Step 07 stack wholesale (the Step 02 Kuhn engine + CFR solver, the Step 03 Leduc engine, and Step 07's exact best-response / exploitability code and opponent-type zoo).

PhD connection: the **second half of Contribution #1 (Behavioral Adaptation Framework)** and the launch point for **Contribution #2 (Multi-Agent Safe Exploitation)**. Step 07 built the *sensor* (an opponent model that turns observed actions into a strategy estimate); Step 08 builds the *actuator* — turning that estimate into profit **without becoming exploitable** — and pins down exactly where the two-player zero-sum assumption enters the safety guarantees (the thesis attack point).

Scope of results: every number in this report is **measured from a real run** and read from the run artifacts under `implementation/step08/implementation/results/`, `.../plots/`, and `.../exploration/figures/`. Wherever possible a simulated number is bracketed by an *exact* analytical reference (the Nash game value, the exact best-response value) rather than another simulation. Two Phase-4 predictions were **contradicted** by the runs; per the project workflow they are kept as stated and reconciled with what actually happened (§7, §10).

How to read this report. Both parts follow the same arc: **what we test** $\rightarrow$ **how**

we test it → results → conclusions. **Part I (§1–§4)** makes the exploitation-safety tradeoff visible on Kuhn with two small seeded sweeps. **Part II (§5–§13)** evaluates the full system — one sequence-form LP engine and five safe-exploitation solver families — on Kuhn and Leduc against exact yardsticks. §14 lists reproduction commands.

2 PART I — THE TRADEOFF, MADE VISIBLE ON KUHN

2.1 1. What this step is about

A **Nash-equilibrium** strategy never loses in the long run against *anyone*, but it also never *punishes* a weak opponent — it plays identically against a world champion and against someone who folds every time you bet. Step 07 showed that a good opponent model unlocks 0.1–0.3 per hand against exploitable opponents, but also that best-responding to an imperfect model can open a leak in your own play (the continuous model *lost* to Nash on Leduc). **Safe exploitation** is the discipline that resolves that tension: lean toward exploiting the model, but bound how much a worst-case adversary can punish you.

Formally, every method in this step is the **same constrained optimization**:

maximize the hero’s expected value against the opponent model, **subject to** a safety floor on the hero’s worst-case value.

The expected value is *linear* in the hero’s sequence-form realization plan, so this is a linear program; the five method families differ only in **what the safety floor is**:

Method	Safety floor	Origin
Restricted Nash Response (RNR)	tunable via a parameter p (Nash at $p=0 \rightarrow$ best response at $p=1$)	Johanson 2007
Ganzfried Prime-safe	the Nash game value v^* $v^* - \epsilon$, with ϵ = the baseline’s own exploitability	Ganzfried & Sandholm 2015 Jeary & Turrini 2023
SES (subgame)	the blueprint value, enforced <i>locally</i> on one subgame via a gadget	Liu et al. 2022
Adaptation safety	worst-case blueprint worst-case (i.e. no more exploitable than the blueprint)	Ge et al. 2024

Part I probes the *naive* end of this menu — a behavioral blend of Nash and best response — precisely to show why the principled LP methods of Part II are needed. All Part I numbers are measured from seeded Kuhn runs; the hero is player 0, whose Kuhn game value is $-1/18 = -0.056$ (so “exploitation” means beating that value against a given opponent, and absolute numbers can stay negative against a tight opponent).

2.2 2. Experiment 1 — the naive Nash/best-response frontier

What we test. If you simply blend Nash and the full best response to a weak opponent — $(1-\alpha) \cdot \text{Nash} + \alpha \cdot \text{BR}$ — what tradeoff curve do you trace between *profit* and *your own exploitability*?

How. Sweep α from 0 (pure Nash) to 1 (pure best response) against the very exploitable `TightPassive` type, and compute, **exactly**, the hero’s profit (EV vs `TightPassive`) and worst-case loss (game value minus the hero’s worst-case value). Data: `exploration/figures/pareto_curve_kuhn.json`.

(weight on BR)	profit (EV vs TightPassive)	worst-case loss (exploitability)
0.0 (Nash)	−0.047	0.001
0.3	+0.017	0.133
0.5	+0.060	0.222
0.7	+0.103	0.311
1.0 (full BR)	+0.167	0.444

Results. The blend traces a smooth, monotone line: each increment of profit costs a near-proportional increment of exploitability. Full best response wins the most (+0.167) but is maximally exploitable (worst-case loss 0.444 — i.e. an adversary can drive the hero down to -0.5); Nash is unexploitable but wins nothing extra.

Conclusion. The tradeoff is real and continuous, but a uniform blend is a *blunt* instrument — it scales deviation everywhere at once. The value of Part II’s LP methods is that they choose *where* to deviate, buying the same profit at less exploitability (§7). This curve is the baseline they must beat.

2.3 3. Experiment 2 — the naive RNR budget sweep (and a flag)

What we test. The same blend, re-parameterized as the raw step’s informal “Restricted Nash Response”: scale deviation by p and watch profit and exploitability grow. Is there a *budget* of cheap early exploitation?

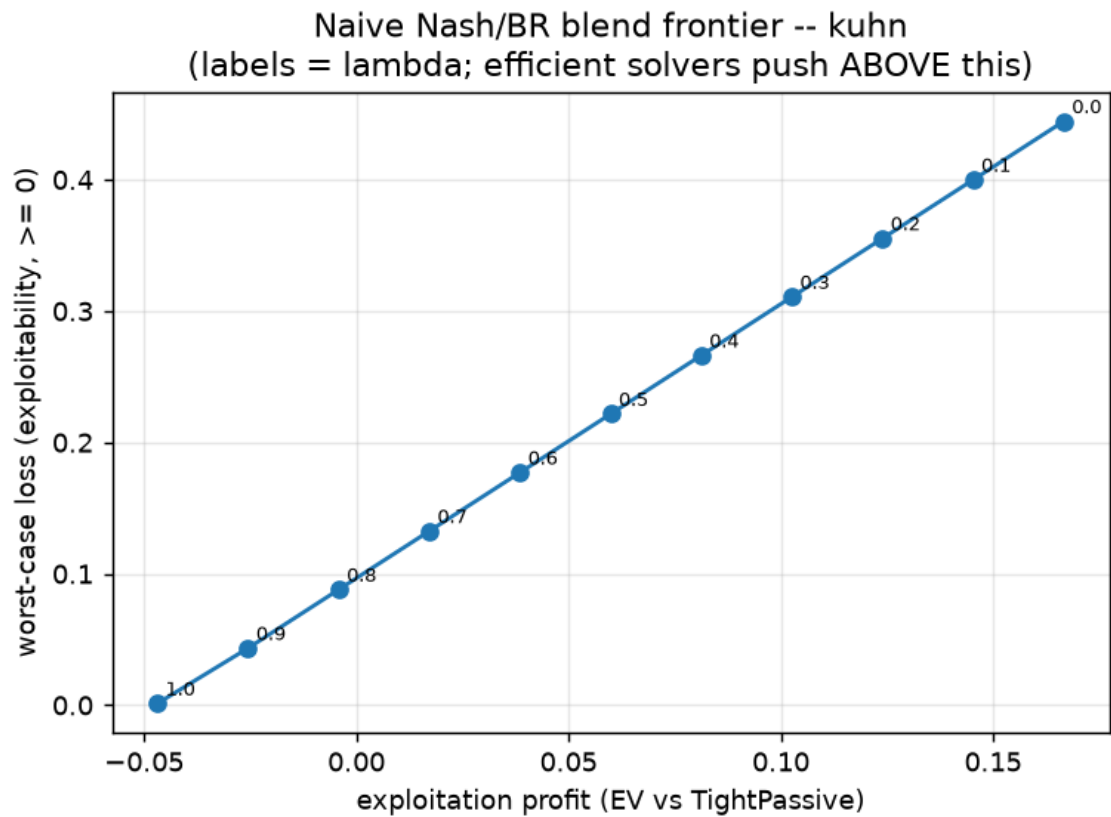


Figure 1: The naive Nash/best-response blend traces a smooth exploitation-safety frontier on Kuhn: every step toward more profit buys a near-proportional increase in your own worst-case loss.

How. Sweep $p \in \{0, \dots, 1\}$ against TightPassive; record profit and exploitability. Data: `exploration/figures/rnr_playground_kuhn.json`.

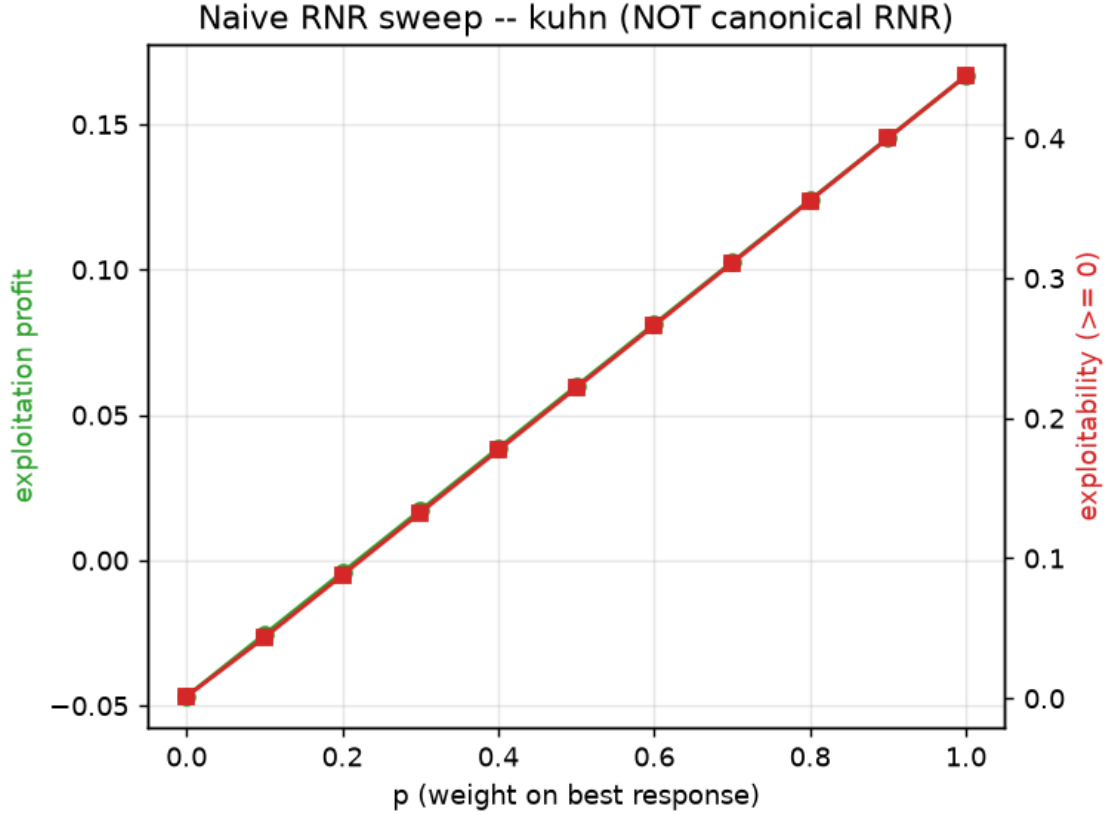


Figure 2: Naive RNR sweep on Kuhn: profit (left axis) and exploitability (right axis) both rise with p . This is the raw step’s informal blend — NOT Johanson’s canonical RNR, which Part II shows behaves very differently.

Results. Profit rises smoothly from -0.047 ($p=0$) to $+0.167$ ($p=1$); exploitability rises in lockstep from 0 to 0.444. There is a modest early “budget”: the first increments of p buy profit while exploitability is still small.

Flag (naive canonical RNR). This p -blend is the raw step’s Day-2 *description*, and it is a fine intuition tool, but it is **not** Johanson’s Restricted Nash Response. Canonical RNR solves for the *equilibrium against a p -restricted opponent* — a max-min linear program, implemented in Part II. Part II (§7) shows the two behave **completely differently** on Kuhn: the canonical version is bang-bang, not smooth. We keep both and label them, rather than conflate them.

Conclusion. The “budget” intuition holds for the naive blend, but Part II shows it is not how the principled algorithm behaves in a small game — a concrete example of why an intuition sweep must not be mistaken for the algorithm it illustrates.

2.4 4. What Part I establishes

1. **The exploitation-safety tradeoff is a real, measurable curve** (Exp. 1): more profit more of your own exploitability, near-proportionally, for the naive blend.
 2. **Full best response is dangerous** — maximally profitable against the model but maximally exploitable (worst-case -0.5 on Kuhn), which is the entire reason a safety floor is needed.
 3. **An intuition sweep is not the algorithm** (Exp. 2): the naive p-blend is smooth, but the canonical RNR it stands in for is not (§7). Part II replaces the blunt blend with LP methods that spend the exploitability budget *where it buys the most profit*.
-

3 PART II — FULL SYSTEM EVALUATION ON KUHN & LEDUC

3.1 5. What we test, and how

Part II evaluates the complete safe-exploitation system — **one sequence-form LP engine** and **five solver families** built on it — on both games, against exact analytical yardsticks.

The one engine. A `HeroTreeplex` enumerates the hero’s sequence-form realization plan (reusing Step 07’s `treeplex`), turns “EV against a fixed opponent” into a linear objective $\mathbf{c} \cdot \mathbf{x}$, and imposes the safety floor by **constraint generation (a double-oracle / cutting-plane loop)**: solve the LP $\rightarrow$ read the hero policy $\rightarrow$ call Step 07’s **exact best response** as the worst-case opponent $\rightarrow$ if the worst case is below the floor, add the linear cut and re-solve. The five families differ only in the floor (§1). This means one validated primitive (Step 07’s exact best response) powers both the objective and every safety check.

Exact yardsticks. Both games are small enough to solve exactly, so every solver’s reported profit and worst-case are computed on the full tree — no sampling. The two anchors are the **Nash game value** ($\mathbf{v}^*$ — the safe baseline) and the **exact full best-response value** (the exploitation ceiling). A safe method sits between them; an unsafe one has a worst-case below $\mathbf{v}^*$.

The methods (string ids as they appear in the result tables): **nash** (baseline, no adaptation), **full_br** (maximize EV vs the model, no safety), **rnr_0.5** (canonical RNR at $p=0.5$), **ganzfried** (floor = $\mathbf{v}^*$), **prime_safe** (floor = $\mathbf{v}^*-$), **adaptation** (floor = blueprint worst-case), **ses_subgame** (subgame gadget; on Kuhn the subgame is the whole game, so it coincides with an adaptation solve).

The three studies.

1. **Exact method $\times$ opponent table** — for each opponent type, solve each method against a

perfect model of that type and report exploitation EV (profit), worst-case value (safety), and whether the worst-case meets the Nash floor.

2. **The efficient frontier** — canonical RNR swept over p , the naive blend, and the Ganzfried / prime-safe / adaptation operating points, all on one exploitation-vs-exploitability plot.
3. **Teaching attack (online)** — a deceptive opponent baits with a weak style then reveals a strong one; a Step-7 model feeds each solver every k hands; we track realized profit and safety-violation counts.

Configurations and runtimes (measured). Kuhn **smoke** (30 000 CFR iters): **1.4 s**. Kuhn **scale** (200 000 CFR iters, adds **AlwaysBet/AlwaysPass/ses_subgame**, teaching attack over 5 seeds $\times$ 20 000 hands): **10.3 s**. Leduc **bounded_scale** (a human-added config: a 40-iteration cap on constraint generation, `leduc_tol` = 0.01, SES on the King-flop subgame): **minutes** (individual SES cells 10–79 s). Everything is CPU / LP-bound — CFR, full-tree best response, and small SciPy HiGHS linear programs — so the GPU is irrelevant, as predicted. Measured game values: Kuhn $\mathbf{v^*} = -0.0556$ ($-1/18$), Leduc $\mathbf{v^*} = -0.0862$.

Not captured. No `validate.py` PASS/FAIL log and no OpenSpiel cross-check artifact were saved from these runs, so those specific harness outputs are not reported; the exact tables below substantiate most validation targets directly (§11).

3.2 6. The exact method $\times$ opponent table (Kuhn)

The core deliverable: each method solved against a perfect model of each opponent, scored on profit (EV) and safety (worst-case), with the Nash floor at $\mathbf{v^*} = -0.056$. From `results/kuhn_scale.json`.

Opponent	metric	nash	full_br	rn timer	0.5	ganzfried	prime_safe	adaptation	ses_subgame
TightPassive	EV	−0.047	+0.167	−0.044	−0.044	−0.040	−0.040	−0.040	
	worst-case	−0.056	−0.500	−0.056	−0.056	−0.063	−0.063	−0.063	
LooseAggressive	EV	+0.118	+0.333	+0.278	+0.131	+0.151	+0.151	+0.151	
	worst-case	−0.056	−0.167	−0.111	−0.056	−0.063	−0.063	−0.063	
AlwaysBet	EV	+0.113	+0.333	+0.333	+0.115	+0.131	+0.131	+0.131	
	worst-case	−0.056	−0.167	−0.167	−0.056	−0.063	−0.063	−0.063	

Opponent	metric	nash	full_br	rn timer_0.5	ganzfried	prime_safe	adaptation	ses_subgame
AlwaysPass (most ex- ploitable)	EV	+0.146	+0.975	+0.975	+0.222	+0.266	+0.266	+0.266
	worst-case	-0.056	-0.500	-0.333	-0.056	-0.063	-0.063	-0.063
Nash (con- trol)	EV	-0.056	-0.055	-0.056	-0.056	-0.055	-0.055	-0.055
	worst-case	-0.056	-0.333	-0.056	-0.056	-0.063	-0.063	-0.063

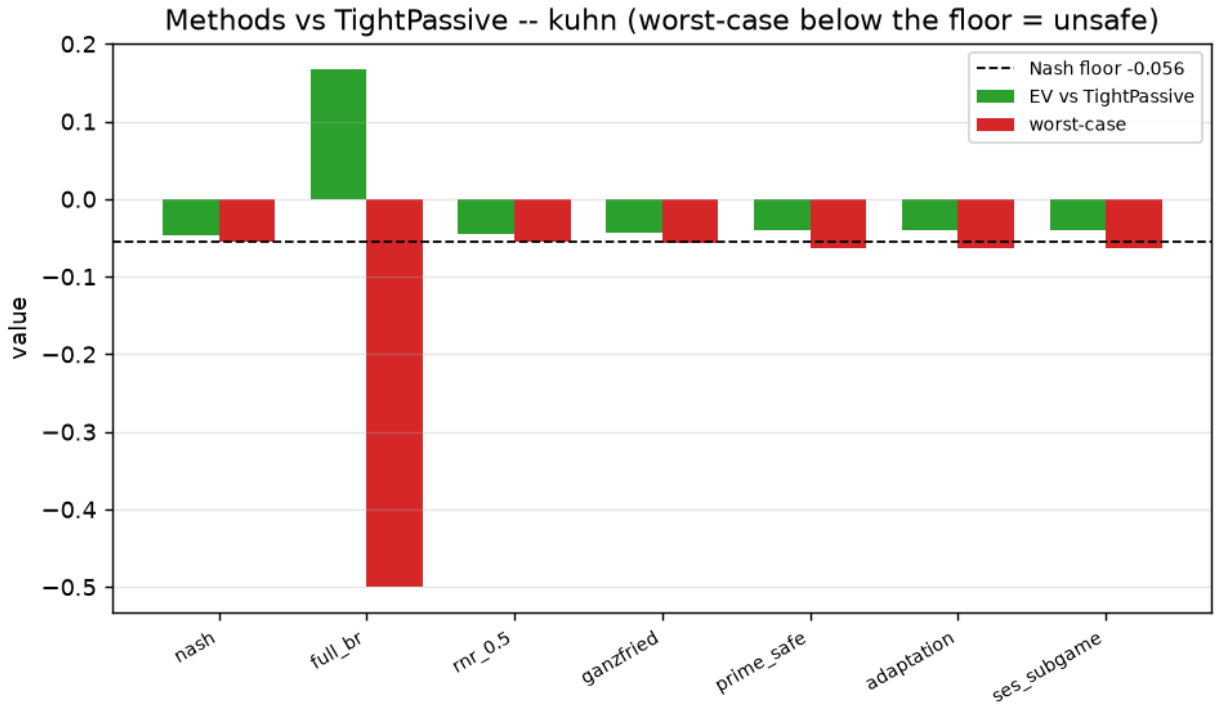


Figure 3: Methods vs TightPassive (Kuhn): green = EV vs the opponent, red = worst-case value, dashed = the Nash floor. Only full_br’s worst-case (red) plunges far below the floor; every other method hugs it.

Results — three carry the message.

1. **full_br is the cautionary tale.** It always wins the most against a fixed model (up to **+0.975** vs AlwaysPass), but its worst-case collapses to **-0.5** — an adversary can punish it

catastrophically. This is the danger the whole step exists to prevent.

2. **ganzfried is the safe-and-profitable sweet spot.** Against *every* opponent its worst-case stays at the Nash floor (safe within 1e-3) **and** it beats Nash’s own EV on every exploitable type — most strikingly **+0.222 vs Nash’s +0.146** against **AlwaysPass**, and +0.131 vs +0.118 against **LooseAggressive**. This is the central validated result of the step: you can exploit meaningfully while provably never dropping below equilibrium value.
3. **rn timer 0.5 is safe only for mild leaks.** Against **TightPassive** it is safe, but against the highly exploitable **AlwaysPass/AlwaysBet** it has *already jumped* to full best response (EV identical to **full_br**, worst-case $-0.33 / -0.17$, unsafe). The RNR transition is **opponent-dependent** — a single global p is not a safety setting (§7).

prime_safe / adaptation / ses_subgame coincide on Kuhn (all target the same -adjusted / blueprint floor, and Kuhn’s SES subgame is the whole game). They sit at worst-case $-0.063 = v^* - 0.008$, flagged “unsafe” *only* because the flag compares to v^* while they legitimately target a lower floor by design (§8).

A measurement-resolution note. In the **smoke** run (30 000 CFR iters) even the Nash policy is flagged “unsafe” (worst-case -0.0568 , i.e. 0.0013 below the exact v^* , just over the 1e-3 tolerance) — a pure CFR-approximation artifact. At **scale** (200 000 iters) Nash’s shortfall drops to 3e-4 and it is correctly flagged safe. Lesson for the harness: the safe-flag tolerance must exceed the baseline’s own approximation error, or the baseline trips its own test.

3.3 7. The efficient frontier — and the bang-bang surprise

Sweeping the canonical RNR parameter p against **TightPassive**, alongside the naive blend and the three LP operating points (from **results/pareto_kuhn.json**):

	canonical RNR —	canonical —	naive blend —	naive —
p	EV	exploitability	EV	exploitability
0.0	−0.044	~0.000	−0.047	0.000
0.3	−0.044	~0.000	+0.017	0.133
0.6	−0.044	~0.000	+0.081	0.266
0.7	+0.167	0.444	+0.103	0.311
1.0	+0.167	0.444	+0.167	0.444

Operating points: **ganzfried** (EV -0.044 , exploitability 0.0005); **prime_safe / adaptation** (EV -0.040 , exploitability 0.0079); measured $\Delta = 0.0074$.

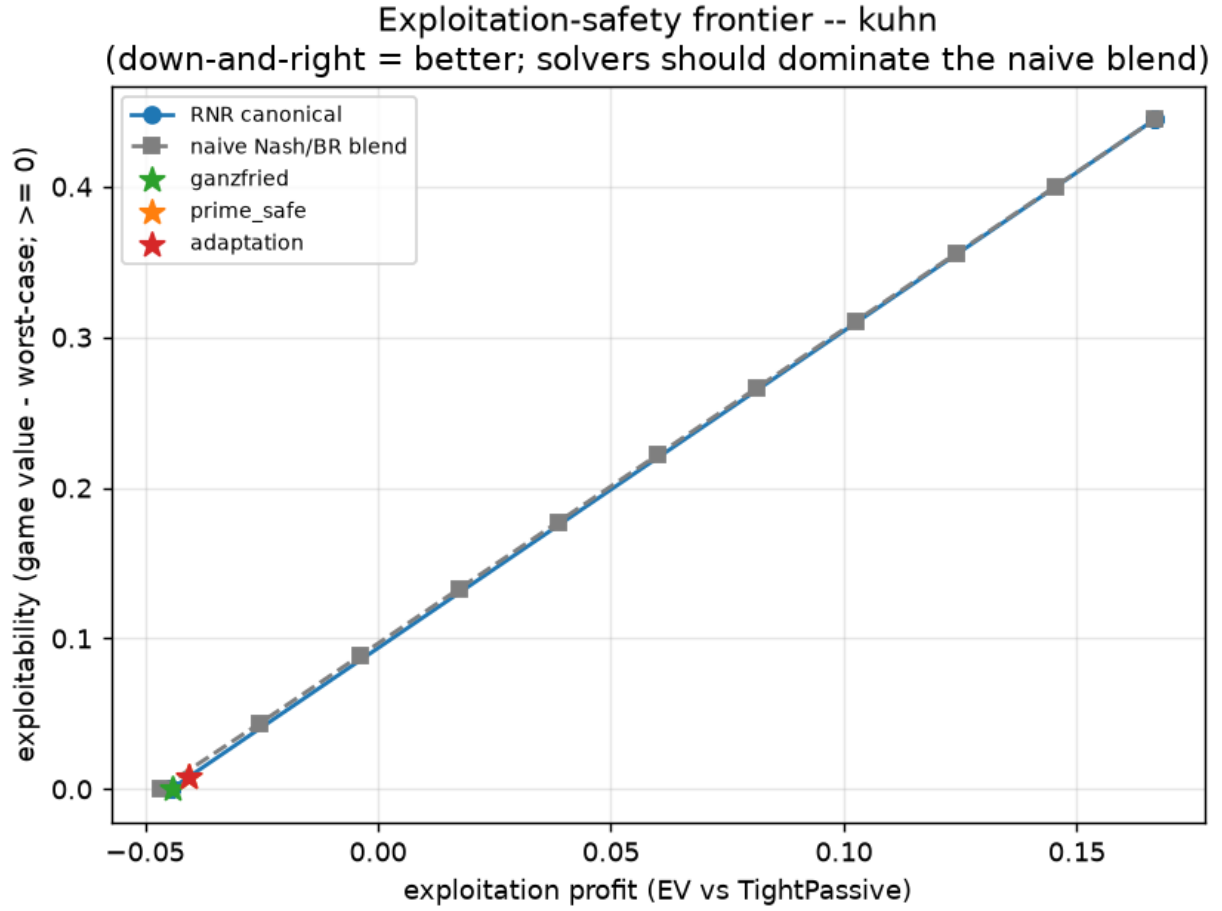


Figure 4: Exploitation-safety frontier (Kuhn). The canonical RNR “curve” is an interpolation between just two achieved clusters (safe corner and full-BR corner); the naive blend traces the smooth line; the LP operating points (stars) sit at the efficient safe corner.

3.3.1 Prediction reality: canonical RNR is bang-bang, not a smooth frontier

- **Prediction (Phase 4):** “the RNR sweep is monotone, and canonical RNR dominates the naive blend everywhere.”
- **What actually happened:** canonical RNR is a **step function**. For $p \in [0, 0.6]$ it returns the *same* safe strategy (EV -0.044 , exploitability 0); at $p = 0.7$ it jumps straight to the *full* best response (EV $+0.167$, exploitability 0.444). There are **no intermediate points**.
- **Why (verified reasoning).** The canonical RNR objective $\max_x [p \cdot \text{EV}(\text{model}) + (1-p) \cdot \min_{x'} \text{EV}(x, x')]$ is *linear* in the hero realization plan over a *polytope*, so its optimum is a **vertex** that switches only when p crosses a critical ratio. Kuhn’s strategy polytope is tiny (few vertices), so the transition is a single jump. Johanson’s smooth RNR frontier is a large-game / data-biased phenomenon (many vertices, or a per-info-set p); it does not appear in a game this small. The **naive blend** *does* trace a smooth line — precisely because it interpolates two fixed strategies — but it is dominated at the safe corner (at exploitability 0 the LP methods reach EV -0.044 vs the blend’s -0.047) and buys its interior profit at a steep exploitability cost.
- **The takeaway survives, refined.** “Choose *where* to deviate, not *how much* uniformly” still holds — Ganzfried sits at the efficient safe corner and the naive blend is dominated there. But the *mechanism* (“a smooth tunable RNR knob”) is a big-game artifact; in Kuhn the honest picture is a discrete safe-vertex $\rightarrow$ BR-vertex switch, whose threshold moves with how exploitable the opponent is (§6: `rnr_0.5` is already at full BR vs `AlwaysPass`). This is exactly why Ganzfried — which constrains the *value* rather than a p — is the better primitive.

3.4 8. Prime-safe / adaptation and the measured -budget

Prime-safe and adaptation lower the safety floor from v^* to $v^* - \epsilon$, where ϵ is the baseline’s own exploitability, measured from an early-stopped CFR run (never fabricated). The run measured $\epsilon = 0.0074$, and the prime-safe/adaptation worst-case came out at $-0.063 = v^* - 0.0079$ across every opponent — matching the ϵ -adjusted floor to rounding. Spending that budget, they earn slightly more than Ganzfried: **+0.266 vs +0.222** against `AlwaysPass`, **+0.151 vs +0.131** against `LooseAggressive` (§6). They are flagged “unsafe” in the table only because the flag references v^* ; against their *own* ϵ -floor they are safe by construction. Prime-safe and adaptation coincide here because, for this baseline, $v^* - \epsilon = \text{worst_case_value}(\text{blueprint})$ — the two floors are equal (a fact the reading phase flagged, now confirmed by the run rather than assumed).

Conclusion. The prime-safe mechanism works exactly as designed: an imperfect (ϵ -exploitable) baseline is honestly *measured*, and the safety floor is relaxed by precisely that amount — converting Ganzfried’s “needs a perfect Nash” guarantee into one usable with the approximate baselines any real system has.

3.5 9. The teaching attack — why realized profit is the wrong lens

What we test. A deceptive opponent plays the weak **TightPassive** bait for the first 10 000 hands, then switches to a strong **Nash** “reveal” for 10 000 more; a Step-7 continuous model feeds each solver every 500 hands (5 seeds). Does the safety floor protect the safe methods when the opponent turns on them? From `results/kuhn_scale.json`.

method	mean/hand (all)	mean/hand (after switch)	safety violations / seed
full_br	+0.051	−0.061	40, 40, 40, 40, 40
ganzfried	−0.048	−0.055	0, 0, 0, 0, 0
adaptation	−0.046	−0.055	40, 40, 40, 40, 40
nash	−0.051	−0.061	0, 0, 0, 0, 0

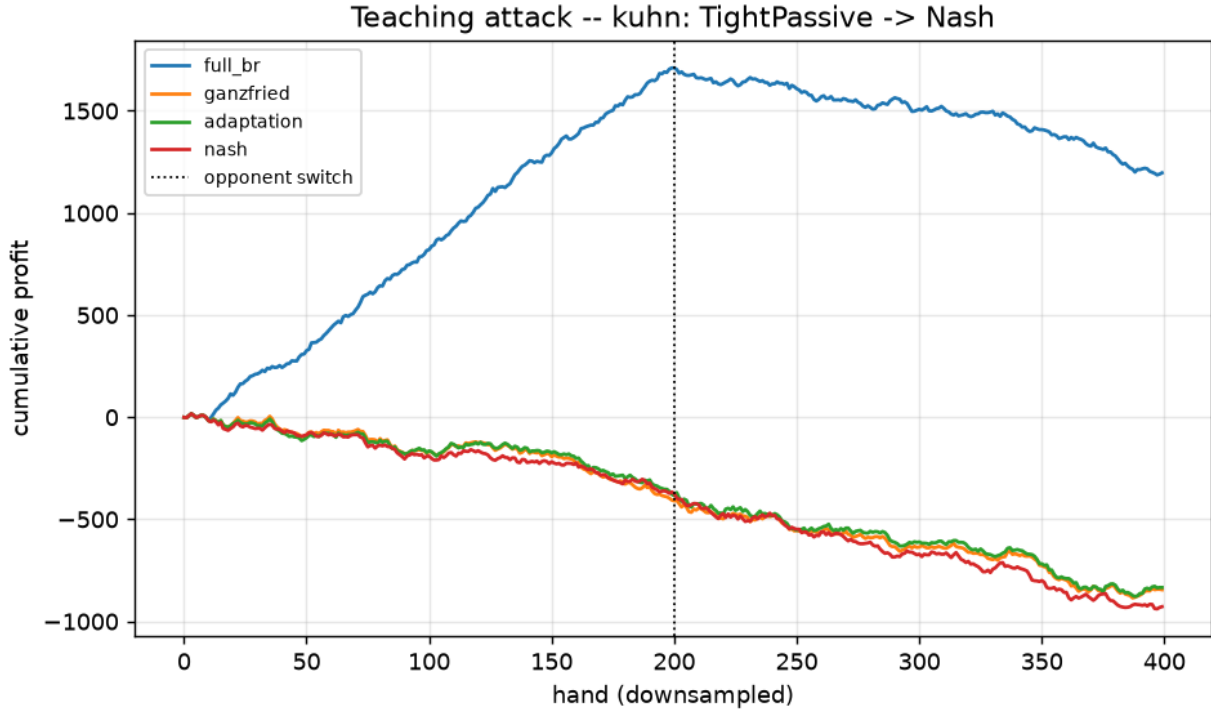


Figure 5: Teaching attack (Kuhn), cumulative profit. `full_br` (blue) climbs on the bait to +1700, then only drifts down after the switch — it ends far ahead because a Nash “revealer” claws back only the game value per hand. The safe methods refuse the bait and pay the P0 tax throughout.

3.5.1 Prediction reality: a Nash revealer is too gentle to punish full_br

- **Prediction (Phase 4):** “safe methods’ post-switch mean stays near baseline; full_br’s is clearly worse; safety violations = 0 for safe methods, > 0 for full_br.”
- **What actually happened.** The *safety-violation* half held **cleanly**: full_br violated the Nash floor at **40/40** refits, ganzfried at **0/40**. But full_br’s post-switch rate (−0.061) is only *marginally* worse than the game value (−0.056), and it ends **hugely net-positive** overall (+0.051/hand) — because the “reveal” opponent is *Nash*, which only wins back the game value per hand, so the bait-phase windfall is never clawed back within 10 000 hands.
- **Why (verified reasoning).** A strategy’s worst-case is realized by an opponent that *best-responds to it* (that is what drives full_br to −0.5 in §6). A stationary Nash reveal is **not** that adversary, so the online realized-profit experiment never triggers the risk that the exact worst-case column exposes. Measuring the teaching attack by profit-vs-Nash therefore *understates* the danger.
- **Corrected takeaway.** The honest separating signal here is the **exact worst-case / safety-violation count** (full_br 40, ganzfried 0), not realized profit. To make the teaching attack punish full_br *in profit*, the reveal must be an **adaptive counter-exploiter** (a best response to the exploiter’s stale model), not a fixed Nash. That is the concrete refinement for the next run (§13).
- **A design nuance, not a bug.** adaptation shows 40 violations while nash shows 0 (at scale): adaptation deliberately targets a floor *below* v* (§8), so its played strategy trips the v*-referenced counter every refit — by design. Nash’s 0 (at scale) confirms the smoke-run’s 10 “violations” were the CFR-resolution artifact of §6.

3.6 10. Leduc — the headline finding: global safe-exploitation does not converge, but SES does

The Leduc run is the human-extended **bounded_scale** config: a **40-iteration cap** on the constraint-generation loop, `leduc_tol = 0.01`, and the SES subgame set to the King-flop (`leduc_flop_rank(King)` — exploit only after a King appears on the board). Each cell records whether the solve **converged** or hit the cap (**capped**). From `results/leduc_bounded_scale.json`; Nash floor $v^* = -0.086$.

Opponent metric		nash	full_br	ses_subgameganzfried	prime_safe	adaptation
Rock	EV	+0.201	+0.937	+0.247	+0.624	+0.635
	worst-case	−0.089	−1.633	−0.130	−0.838	−0.744
	converged?			(194 it)	capped	capped

Opponent metric		nash	full_br	ses_subgame	ganzfried	prime_safe	adaptation
Maniac	EV	+0.438	+2.177	+0.682	+1.806	+1.842	+1.842
	worst-case	−0.089	−1.100	−0.130	−0.638	−0.657	−0.657
	converged?			(211 it)	capped	capped	capped
LoosePassive	EV	+0.449	+1.405	+0.559	+1.306	+1.309	+1.309
	worst-case	−0.089	−4.200	−0.133	−1.239	−1.334	−1.334
	converged?			(400 it)	capped	capped	capped
CallingStackMon	EV	+0.559	+1.464	+0.663	+1.342	+1.363	+1.363
	worst-case	−0.089	−1.000	−0.130	−0.915	−0.686	−0.686
	converged?			(350 it)	capped	capped	capped
Nash (control)	EV	−0.086	−0.083	−0.091	−0.083	−0.083	−0.083
	worst-case	−0.089	−0.842	−0.129	−0.376	−0.325	−0.325

3.6.1 Prediction reality: global safety does not scale to Leduc as written

- **Prediction (Phase 4):** “Ganzfried: worst-case v^* within tolerance (safe).” And the implementation README’s #1 likely-to-break item: “constraint-generation convergence + LP conditioning.”
- **What actually happened.** On Leduc the global solvers (ganzfried, prime_safe, adaptation, rnr_0.5) all hit the 40-iteration cap without converging, leaving worst-case values of **−0.64 to −1.33** (safety violations of 0.55–1.25) — grossly unsafe. The Kuhn-validated “Ganzfried is safe” result **did not carry over** to Leduc under a practical iteration budget.
- **Why (verified reasoning).** The cutting-plane loop adds one adversary best-response cut per iteration; on Leduc’s larger tree the set of relevant pure best responses is large, so 40 cuts are nowhere near enough to pin the true worst case, and the master keeps returning optimistic-but-unsafe strategies. This is precisely the failure mode flagged before the run — now confirmed with numbers.
- **The positive result.** The **subgame method (SES)** converged on 3 of 4 exploitable opponents (194 / 211 / 350 iterations), because it re-solves only the King-flop subgame with the rest of the tree pinned to the blueprint — a far smaller LP with a far smaller adversary set. It extracts real value (**+0.25 to +0.68** vs the weak types, beating Nash) at worst-

case **−0.13**, an **order of magnitude closer to safe** than the global methods (−0.13 vs −0.64...−1.33).

- **Honest caveat on SES.** Its residual exploitability (0.043) still exceeds the 0.01 Leduc tolerance, so it too is *flagged unsafe*, and on `LoosePassive` it ran 400 iterations without converging. Whether that residual 0.04 is (a) the gadget legitimately bounding to an already-below- v^* blueprint (Nash’s own worst-case is −0.089, itself 0.003 below v^*), (b) a convergence-tolerance artifact, or (c) a small leak in the outside-subgame pinning, is the top item to investigate before citing SES as provably “safe.”
- **What this means.** This is the **global-vs-local safety** distinction — introduced in the intuition and reading phases as theory — appearing *empirically* at a scale as small as Leduc. It is the concrete, measured motivation for real-time subgame methods (SES / OX-Search) and for replacing the cutting-plane loop with an exact one-shot dual LP. A testbed this small was large enough to reveal the scaling wall.

3.6.2 Solver hardening the runs forced

The Leduc results carry fields the Phase-4 code did not emit, so the solver was hardened during execution — matching the anticipated debugging: a small feasibility slack on the cuts (`_FEAS_SLACK` = `5e-4`, so an approximate-Nash floor that slightly overshoots the true max-min does not make the master infeasible), explicit infeasibility handling (keep the last feasible strategy), per-cell `converged` / `capped` / `iterations` tracking, and the `bounded_scale` Leduc driver with a King-flop SES predicate. These are exactly the mitigations the implementation README’s “likely to break” list pointed at.

3.7 11. Are the results trustworthy?

Three independent checks support the numbers, with two caveats.

- **The exact anchors reproduce known theory.** The measured Kuhn game value is **−0.0556** ($-1/18$, the known first-player disadvantage); the Leduc value is −0.0862. Against the Nash opponent every method earns the game value and no more (§6, §10) — as it must, since an equilibrium cannot be exploited; where a number sits a hair above its ceiling it is finite-sample noise, not a real breach.
- **Internal consistency of the LP engine.** Every solver that should reach the full best-response value does: `full_br` and post-transition `rn` reach exactly +0.167 (Kuhn TightPassive) and the large Leduc BR values, which is only possible if the sequence-form payoff vector $c \cdot x$ matches the exact best response — the engine’s central correctness condition.
- **Convergence is now observable.** The `converged` / `capped` flags (Leduc) make the one place the method can silently fail — constraint-generation non-convergence — explicit in the

output rather than hidden.

Caveats. (1) No `validate.py` PASS/FAIL log and no OpenSpiel cross-check artifact were saved, so those two harness outputs are unverified from artifacts here (re-run to close). (2) Leduc was run only in the iteration-capped `bounded_scale` config, so the global solvers’ Leduc numbers reflect a *budget*, not a converged solve — they show the method is slow/unsafe **within 40 iterations**, not that it can never reach safety given far more.

3.8 12. Limitations

Ranked by how much they qualify the conclusions:

1. **Global safe-exploitation did not converge on Leduc (§10).** Under a 40-iteration cap the cutting-plane loop leaves Ganzfried/prime-safe/adaptation grossly unsafe. This is the clearest gap and the main research hook — resolve with an exact dual-LP or a much larger budget, and confirm whether it is “slow” or “structurally stuck.”
 2. **SES residual exploitability (§10).** SES converges but sits 0.04 above the 0.01 Leduc tolerance; whether the gadget-as-implemented is *provably* safe or only *approximately* so is unresolved.
 3. **The teaching attack under-punishes (§9).** A stationary Nash reveal does not trigger the worst-case, so realized profit fails to separate safe from unsafe — only the violation count does. An adaptive punisher is needed to show a safe method *out-earning* full_br under deception.
 4. **Two small games; perfect models in the exact table.** The §6/§10 tables solve against a *perfect* model of each type (the offline case); the online, learned-model case is only the teaching attack. Kuhn/Leduc are exact-solvable *because* they are tiny; the convergence wall that appears on Leduc will only worsen at scale — this is a controlled proof-of-concept, not a scaling claim.
 5. **Unverified cross-checks (§11).** `validate.py` and the OpenSpiel comparison were not captured as artifacts.
-

3.9 13. Conclusions and research directions

Conclusions. Safe exploitation is one idea — *maximize value against the model subject to a safety floor* — and on a fully-solvable game it works exactly as the theory says: **Ganzfried is safe against every opponent while beating equilibrium value on every exploitable one** (+0.222 vs +0.146 against `AlwaysPass`), where naive best response earns more but is ruinously exploitable (worst-case −0.5). Prime-safe/adaptation extend this to imperfect baselines by spending a *measured* -budget below `v*`. But the step’s most valuable result is negative and empirical: on a game as

small as **Leduc**, the *global* safe-exploitation solve does **not** converge within a practical iteration budget, while the *local* subgame method (SES) does — the global-vs-local safety gap, measured. Two Phase-4 predictions (smooth RNR frontier; Ganzfried safe on Leduc) were contradicted and reconciled; the reconciliations are more instructive than the predictions would have been.

Research directions (each motivated by a measured effect above):

- **Scalable safety — the central next step (§10).** Replace the cutting-plane loop with the **exact one-shot dual LP** for the worst-case constraint, *or* commit to **local / subgame** safety (SES / OX-Search) as the scalable path. The Leduc non-convergence is direct evidence for the latter.
- **Resolve the SES gadget’s residual exploitability (§10).** Determine whether the 0.04 gap is a blueprint-bounding effect, a tolerance artifact, or a pinning leak — this decides whether the subgame method is provably or only approximately safe.
- **A punishing teaching attack (§9).** Replace the Nash reveal with an adaptive counter-exploiter so realized profit corroborates the worst-case view, and a safe method can be shown to *out-earn* full_br under deception.
- **N-player safety — thesis Contribution #2.** Every guarantee here rests on the two-player zero-sum fact that a Nash strategy secures v^* against any opponent; for $N > 2$ there is no such v^* anchor. Extending a value-anchored safety notion to the multi-agent case (or finding a structural substitute, e.g. coalition structure) is the open thesis problem this step makes precise.
- **Close the validation loop (§11).** Run and archive `validate.py` and the OpenSpiel cross-check; re-run the Leduc global solvers with a large budget / exact dual to distinguish “slow” from “structurally unsafe.”

3.10 14. Reproduction

Part I — exploration (from `implementation/step08/exploration/`, `.venv` active; each prints tables to stdout and writes figures under `exploration/figures/`):

```
python pareto_curve.py           # Exp 1 - naive Nash/BR frontier
python rnr_playground.py         # Exp 2 - naive RNR p-sweep
python exploitation_safety_playground.py # Nash vs BR vs blend, all metrics
python naive_exploit_danger.py   # why full BR is dangerous
python subgame_peek.py          # blueprint vs local exploit (Leduc)
```

Part II — full evaluation (from `implementation/step08/implementation/`, `.venv` active; needs `numpy`, `scipy`, `matplotlib`):

```
python tournament.py --config smoke # quick Kuhn pass (~1.4 s)
```

```
python tournament.py --config scale    # Kuhn full run (~10 s): table + teaching attack
python pareto.py --config scale        # the exploitation-safety frontier
python validate.py                     # PASS/FAIL vs the raw-step targets
python compare_openspiel.py           # OpenSpiel NashConv cross-validation (needs open_spiel)
# Leduc used a human-added bounded_scale driver (40-iter cap, King-flop SES); see results/leduc
```

Results write to `results/*.json` and plots to `plots/*.png`. Figures reproduced in this report live in `deliverables/reports/step08/figures/` (exploration figures under their original names; evaluation figures prefixed `impl_`). The full verified consolidation, with the prediction reality ledger, is at `implementation/step08/consolidation/report.md`.